

Analysis 1: Aug-Nov 2026

Rule: While I encourage you all to read other books and sources, while writing proofs try to stick to the matters we have discussed in class or in exercises. Important definitions used in class I will include up in exercises. While writing answers to be looked at by TAs now (and the same rule for mid-sem/end-sem) if you use any result other than the ones mentioned here - you have to write the full statement along with the proof. It is not enough to give name of a result or say result abc from book by author xyz!

Definition: Let $B \subseteq \mathbb{R}$. $a \in \mathbb{R}$ is said to be the least upper bound (written as *l.u.b.*) of B if (i) $x \leq a \forall x \in B$ and (ii) if $d \in \mathbb{R}$, $x \leq d \forall x \in A$ then $a \leq d$.

The *l.u.b.* Axiom for real numbers is If $B \subseteq \mathbb{R}$ admits an upper bound, then B also admits a *l.u.b.*.

We will be assuming throughout this course that $\mathbb{R}$ satisfies *l.u.b.* Axiom.

Definition: Let $B \subseteq \mathbb{R}$. $a \in \mathbb{R}$ is said to be the greatest lower bound (written as *g.l.b.*) of B if (i) $x \geq a \forall x \in B$ and (ii) if $d \in \mathbb{R}$, $x \geq d \forall x \in A$ then $a \geq d$.

Exercises 1

Here $\langle a_n : n \geq 1 \rangle$, will denote a sequence of real numbers. Can be abbreviated as $\langle a_n \rangle$.

Definition: A sequence $\langle a_n : n \geq 1 \rangle$ is said to **converge** to a if $\forall \epsilon > 0$, $\exists n^*$ such that $m \geq n^*$ implies $|a_m - a| < \epsilon$.

Definition: A sequence $\langle a_n : n \geq 1 \rangle$ is said to be **Cauchy** if $\forall \epsilon > 0$, $\exists n^*$ such that $n, m \geq n^*$ implies $|a_n - a_m| < \epsilon$. Let $\langle a_n : n \geq 1 \rangle$ be a Cauchy sequence.

(1) Suppose $\langle a_n : n \geq 1 \rangle$ is a monotonically increasing bounded sequence of real numbers. Show that $\exists a \in \mathbb{R}$ such that $\lim_{n \rightarrow \infty} a_n = a$ (also written as $a_n \rightarrow a$).

Hint: Let B denote the set $\langle a_n : n \geq 1 \rangle$ and take $a = \text{l.u.b. } B$.

(2) Let $\langle a_n : n \geq 1 \rangle$, $\langle b_n : n \geq 1 \rangle$ be convergent sequences. Let $c_n = a_n + b_n$, $d_n = a_n b_n$, $\forall n \geq 1$. Then show that $\langle c_n : n \geq 1 \rangle$, $\langle d_n : n \geq 1 \rangle$ are also convergence.

(3) Let $\langle a_n : n \geq 1 \rangle$, $\langle b_n : n \geq 1 \rangle$ be convergent sequences. Suppose $\lim_{n \rightarrow \infty} a_n = a$ and $\lim_{n \rightarrow \infty} b_n = a$. Suppose $\langle c_n : n \geq 1 \rangle$ is such that $a_n \leq c_n \leq b_n$, $\forall n \geq 1$. Show that $\lim_{n \rightarrow \infty} c_n = a$.

(4) Let $\langle a_n : n \geq 1 \rangle$, $\langle b_n : n \geq 1 \rangle$ be convergent sequences with $\lim_{n \rightarrow \infty} a_n = a$, $\lim_{n \rightarrow \infty} b_n = b$, with $b \neq 0$. For $n \geq 1$, let $u_n = 0$ if $b_n = 0$ and $u_n = \frac{a_n}{b_n}$ if $b_n \neq 0$. Show that $\langle u_n : n \geq 1 \rangle$ is convergent.

The sequence $\langle u_n : n \geq 1 \rangle$ as defined above is also expressed as $u_n = \frac{a_n}{b_n} \mathbf{1}_{\langle b_n \neq 0 \rangle}$.

(5) Let $\langle a_n : n \geq 1 \rangle$ be a convergent sequence. Show that it is a Cauchy sequence.

(6) Let $\langle a_n : n \geq 1 \rangle$ be a Cauchy sequence. Show that

[a] $\langle a_n : n \geq 1 \rangle$ is bounded.

[b] For $n \geq 1$ let E_n be the set defined by $E_n = \{a_k : k \geq n\}$ and let $t_n = \text{l.u.b. } E_n$.

Show that $\langle t_n : n \geq 1 \rangle$ is a bounded monotonically decreasing sequence.

[c] Show that $\langle t_n : n \geq 1 \rangle$ is a convergent sequence.

[d] Let $a = \lim_{n \rightarrow \infty} t_n$. Show that $\langle a_n : n \geq 1 \rangle$ converges to a .

(7) Let $\langle a_n : n \geq 1 \rangle$ be a bounded sequence. For $n \geq 1$ let E_n be the set defined by $E_n = \{a_k : k \geq n\}$ and let $t_n = \text{l.u.b. } E_n$, $s_n = \text{g.l.b. } E_n$. Show that

[a] For all $n \geq 1$, $s_n \leq s_{n+1} \leq t_{n+1} \leq t_n$

[b] $\lim_{n \rightarrow \infty} s_n = s$, $\lim_{n \rightarrow \infty} t_n = t$ exist and $s \leq t$.

[c] $\langle a_n : n \geq 1 \rangle$ is Cauchy if and only if $s = t$.

[c] $\langle a_n : n \geq 1 \rangle$ converges to a if and only if $s = t = a$.

The quantities s , t defined in [b] above are respectively called $\liminf$ of $\langle a_n : n \geq 1 \rangle$ and $\limsup$ of $\langle a_n : n \geq 1 \rangle$, and expressed as $s = \liminf_{n \rightarrow \infty} a_n$, $t = \limsup_{n \rightarrow \infty} a_n$.